



A survey on maritime fleet size and mix problems



Giovanni Pantuso^{a,*}, Kjetil Fagerholt^{a,b}, Lars Magnus Hvattum^a

^a Norwegian University of Science and Technology, Alfred Getz vei 3, 7491 Trondheim, Norway

^b Norwegian Marine Technology Research Institute (MARINTEK), Otto Nielsens vei 10, 7052 Trondheim, Norway

ARTICLE INFO

Article history:

Available online 9 May 2013

Keywords:

Logistics
Maritime transportation
Fleet planning

ABSTRACT

This paper presents a literature survey on the fleet size and mix problem in maritime transportation. Fluctuations in the shipping market and frequent mismatches between fleet capacities and demands highlight the relevance of the problem and call for more accurate decision support. After analyzing the available scientific literature on the problem and its variants and extensions, we summarize the state of the art and highlight the main contributions of past research. Furthermore, by identifying important real life aspects of the problem which past research has failed to capture, we uncover the main areas where more research will be needed.

© 2013 Elsevier B.V. All rights reserved.

1. Introduction

As the first decade of the third millennium ended with a worldwide financial and economic crisis, several countries experienced a decrease in their gross domestic product. In 2009 the world trade fell by 22.9% compared with the year before, the deepest fall in 70 years, and this strong downturn also affected seaborne volumes, which decreased by 4.5% (UNCTAD, 2010). Looking beyond the recent downturn, maritime economics has always been characterized by a cyclic repetition of peaks and troughs in demand and freight rates. While the demand of maritime transportation reacts quickly to changes in freight rates, the supply adapts slowly to changes in demand, mostly because of the long lead time associated with the acquisition of new ships. Imbalances between supply and demand are therefore common. This can be illustrated by the fact that in 2009 the world fleet grew by 7% over the year before, a growth that continued also in the beginning of 2010, despite the reduction in trade volumes. The tonnage oversupply was the result of orders for new ships submitted before the downturn. One often sees that in a trough, the tonnage is renewed, while in a peak one tends to postpone the demolition of older ships. Illustrating this pattern, a 300% increase in demolitions of old tonnage was observed during 2009 (UNCTAD, 2010).

These dynamics are one of the main triggers of the wave-motions in freight rates which are typically referred to as shipping market cycles. They can be described as the overlapping of three different cycles (Stopford, 2009):

1. long-term cycles, typically driven by major changes in the industries of seaborne commodities,

2. short-term cycles, which mainly follow the evolution of the world economy, and
3. seasonal cycles, characteristic of many seaborne commodity trades (e.g. agricultural ones).

Shipping companies operate in such an uncertain and changeable environment, and a crucial strategic decision is that of designing an optimal fleet of ships. In its basic version the *maritime fleet size and mix problem* (MFSMP) consists of deciding how many ships of each type to use in order to meet the demand. The objective is typically to minimize the total cost of setting up and operating a fleet of ships and usually the problem includes ship routing or deployment decisions to support the tonnage estimation.

An example of an objective function for a basic version of the MFSMP is given in (1), and consists of a fixed term associated with the acquisition of ships and a variable term associated with their operations.

$$\min \sum_{v \in V} C_v^F y_v + \sum_{v \in V} \sum_{r \in R_v} C_{vr}^V x_{vr} \quad (1)$$

Here, V is the set of available ship types and R_v represents the set of routes r that a ship of type v can sail. In the first term of (1) C_v^F represents the cost of including a ship of type v in the fleet, while variable y_v represents the number of ships of type v to include. In the second term, C_{vr}^V represents the cost of sailing route r with ships of type v and decision variable x_{vr} represents the number of times route r is sailed by ships of type v .

To ensure feasibility of the fleet operations, constraints need to keep track of the consumption of resources associated with the ships. Constraints (2) provide an example, where Z_{vr} is the time consumed every time a ship of type v sails route r , and Z represents the total amount of time available for each ship within the planning horizon. In some applications, other resources than time, such

* Corresponding author. Tel.: +47 45196397.

E-mail address: pantuso@iot.ntnu.no (G. Pantuso).

as the number of available ships of a type, may be modeled in a similar way.

$$\sum_{r \in R_v} Z_{vr} x_{vr} - Z y_v \leq 0, \quad v \in V \quad (2)$$

Additionally, constraints are needed to ensure that ship operations are performed to meet the demand. Constraints (3) represent an example of such constraints ensuring that each port (or region) $i \in N$ is called at least D_i times during the planning horizon. The parameter A_{ir} is equal to 1 if route r calls port i , and is equal to 0 otherwise. Ships have to sail each route a number of times sufficient to meet the frequency requirement for each port. Alternatively, or in addition to constraints (3), one may want to control the amount of cargo shipped to each port. In this case, let D_i be the demand of port i and Q_v the capacity of a ship of type v . Constraints (3) could then be modified by multiplying the argument of the summation in the left-hand side by parameter Q_v . The left-hand side would then represent the total amount of cargo shipped to port i . However, the type of ship operations is very much dependent on the specific problem and may vary substantially from one problem to another. Therefore, different or additional restrictions might be wanted.

$$\sum_{v \in V} \sum_{r \in R_v} A_{ir} x_{vr} \geq D_i, \quad i \in N \quad (3)$$

Finally, a typical mathematical model includes restrictions on the variables domain, where variables of type y_v are usually restricted to take integer values while the restrictions put on variables of type x_{vr} depend on the specific problem.

In this paper we present a literature survey on the MFSMP and its variants and extensions. Based on the survey, we discuss the state of the art and point out possible directions for future research within the subject. The remainder of this paper is organized as follows. In Section 2 we give a more thorough motivation for the survey. The relevant literature is examined in Section 3. In Section 4 the state of the art is discussed and future research perspectives are pointed out. Concluding remarks are given in Section 5.

2. Background and motivation for the survey

A survey on the general fleet size and mix problem was presented by Hoff et al. (2010), who discussed the industrial aspects of the fleet composition and routing. The main focus was on the fleet size and mix vehicle routing problem and its variants. However, methods proposed for the fleet size and mix vehicle routing problem are not necessarily applicable to maritime problems which in most cases have operational characteristics that differ from those implied by a vehicle routing problem structure (see the discussions by Ronen (1983) and Christiansen et al. (2007)). Hoff et al. (2010) also surveyed a number of applications of both land-based and maritime problems. However, additional studies describing maritime applications are available in the literature, besides the ones they listed.

A specific investigation of the maritime applications literature is in place because several aspects of the MFSMP, other than the operational differences, make it different from the fleet size and mix problem for other transportation contexts. Distinguishing elements are, for example: (1) higher level of uncertainty, (2) higher amount of capital involved, and (3) the ships' value function. Below we elaborate upon each of these characteristics for maritime applications.

Uncertainty in maritime transportation, as discussed in the introduction, affects all planning levels. To the best of our knowledge, no other transportation context is affected by such high level of uncertainty in demand, ship costs and freight rates. As an exam-

ple, in 2009 the average daily charter rate for 1600–1999 TEUs container ships fell by 67.6% from 2008, and in 2010 it was less than half than in 2008 (UNCTAD, 2011). Furthermore, uncertainty is emphasized by the long lifetime of ships which is usually around 30 years. This is much longer than for road-based vehicles, but can be comparable to the lifetime of aircraft and trains. Therefore, investments in ships require taking a long-term view of the shipping company's prospects. To this extent, Stopford (2009) suggests that the direction of change for the geopolitical environment should be the starting-point for any future analysis, rather than economics.

The amount of capital needed to acquire new (or second-hand) ships also distinguishes MFSMPs from their land-based counterparts, and is comparable with the amounts needed to acquire new airplanes. New ships may cost up to hundreds of million USD and this increases the relevance of the financing of the investment. Generally, several financing alternatives are available and the chosen one will influence the capital cost of a ship (i.e. the sum of debt repayment and interest or dividend). These costs can amount to more than 40% of the total running costs even for a ten-year-old ship (Stopford, 2009).

Finally, the evolution of the value (and price) of ships differ from that of most other vehicles which usually decreases as time goes by. As an example, Couillard and Martel (1990) modeled the value of road vehicles as a decreasing function of age and mileage. The value of a ship is a more complex parameter to model. Adland and Koekebakker (2007) conclude that the second-hand value of a given type of ship can be described as a non-linear function of three parameters: size, age, and the state of the freight market. Several other studies exist on modeling the variation of ship value over time. Examples are Tsolakis et al. (2003) and Adland and Koekebakker (2004).

In this paper we focus on MFSMPs for shipping companies that are primarily interested in ships for transportation purposes. That is, we will not include literature where the MFSMP has been studied under the perspective of asset play (see, e.g., Alizadeh and Nomikos, 2007; Marcus et al., 1991; Bendall and Stent, 2005 and Sodal et al., 2009).

3. Literature review

In the following literature review on the MFSMP we distinguish between single-period MFSMPs which will be referred to simply as MFSMPs, and multi-period MFSMPs which will be referred to as *maritime fleet renewal problems* (MFRP). The former category of problems focus on the design of a fleet of ships for transportation systems whose characteristics are meant to remain unchanged over time and therefore do not take into account the evolution from a point in time to another. They may also represent short-term operations. The latter is an extension of the MFSMP in which a dynamic adjustment of the fleet in response to the evolution of the service requirements is sought. In this case the problem implies an existing fleet to be renewed from time to time and a planning horizon which is to be considered a succession of time periods. A variant of the MFSMP is the *maritime fleet size problem* (MFSP) which consists of determining the number of ships in a homogenous fleet, i.e. where all the ships have identical characteristics.

For each of the papers reviewed, Tables 2–5 report the following information: the way ships are acquired or disposed of, the mode of operations, the industry the study deals with, the methodology applied, and the type of operating decision the tonnage estimation is based on. As far as the operations mode is regarded, we refer to the classification given by Lawrence (1972), which distinguishes between tramp, liner and industrial. Tramp shipping operators

Table 1

Notation used in the following tables.

Acquisition/disposal of	Mode	Methodology	Operating decision
CI = Charter in CO = Charter out P = Purchase B = Building CP = Choose from a pool LU = Lay up SE = Sale SC = Scrapping – = Not reported	LI = Liner TR = Tramp IN = Industrial	HEU = Heuristic StP = Stochastic programming SIM = Simulation QM = Queue modeling NLP = Non-linear programming OC = Optimal control RO = Robust optimization	D = Deployment R = Routing S = Scheduling – = Not reported

Table 2

Papers on the strategic MFSMP.

Paper	Acquisition/disposal	Mode	Industry	Methodology	Operating decisions
Bradley et al. (1977)	P	IN	Bulk shipping	LP	D
Crary et al. (2002)	P	IN	Navy	MIP	D
Darzentas and Spyrou (1996)	–	LI	Passengers transportation	SIM	–
Everett et al. (1972)	P	IN	Bulk shipping	LP	D
Fagerholt et al. (2010)	–	TR or IN	General shipping	SIM	R + S
Larson (1988)	–	IN	Refuse transportation	HEU	D
Murotsu and Taguchi (1975)	B	TR or IN	Oil shipping	DP + NLP	–
Pesenti (1995)	P	TR	Container shipping	MIP + HEU	D
Sigurd et al. (2005)	P	LI	Container shipping	MIP + HEU	R + S
Zeng and Yang (2007)	–	TR	Coal shipping	IP + HEU	R

Table 3

Papers on the short-term MFSMP.

Paper	Acquisition/disposal	Mode	Industry	Methodology	Operating decisions
Fagerholt (1999)	CI	LI	General shipping	IP	R
Fagerholt and Lindstad (2000)	CI	IN	Offshore services	IP	R + S
Halvorsen-Weare and Fagerholt (2011)	CI	IN	Offshore services	IP + SIM	R
Halvorsen-Weare et al. (2012)	CI	IN	Offshore services	IP + HEU	R
Lane et al. (1987)	CP	LI	General shipping	IP	R + S
Mehrez et al. (1995)	CI	IN	Bulk shipping	MIP + HEU	R
Meng and Wang (2010)	CP,CI,CO	LI	Container shipping	StP	D
Meng et al. (2012)	CP,CI,CO	LI	Container shipping	StP	D
Schwartz (1968)	CP,CI	TR	River system	MIP	R + S

operate on customers' callings. In addition to cargoes whose transportation is agreed through long-term contracts, they typically try to increase the profit by carrying optional spot cargoes. In liner shipping ships sail according to a public fixed schedule. Their profits are therefore influenced by the schedule and frequency published. Finally, in the industrial mode the owner of the goods to transport operates its own ships and seeks to minimize transportation cost. Table 1 reports the notation and the abbreviations, other than those introduced when needed throughout the text, used in the tables.

Section 3.1 describes the available papers on the MFSMP. Papers on the MFSP are presented in Section 3.2, while Section 3.3 reports the available papers on the MFRP. Finally, Section 3.4 presents some papers in which the fleet size and mix is corrected within the frame of tactical problems.

3.1. Maritime fleet size and mix problems

In this section we first present the available studies on the MFSMP as a strategic problem, and second studies that consider

Table 4

Papers on the MFSP.

Paper	Acquisition/disposal	Mode	Industry	Methodology	Operating decisions
Bellmore et al. (1968)	–	Li or IN	Oil shipping	LP	S
Bendall and Stent (2001)	–	TR, Li or IN	Container shipping	MIP	S
Dantzig and Fulkerson (1954)	–	Li or IN	Oil shipping	LP	S
Jaikumar and Solomon (1987)	–	Li	River system	MIP	R
Koenigsberg and Lam (1976)	–	TR, Li or IN	LNG shipping	QM + SIM	–
Koenigsberg and Meyers (1980)	–	TR, Li or IN	LNG shipping	QM + SIM	–
Lai and Lo (2004)	CI, P	Li	Passengers transportation	MIP + HEU	R + S
Larson et al. (1991)	P	IN	Refuse transportation	SIM	R
Richetta and Larson (1997)	P	IN	Refuse transportation	SIM	R
Sambracos et al. (2003)	–	Li	Container shipping	MIP	R
Shyshou et al. (2010)	CI	IN	Offshore services	SIM	–

Table 5

Papers on the MFRP.

Paper	Acquisition/disposal	Mode	Industry	Methodology	Operating decisions
Alvarez et al. (2011)	P, CI, CO, SE, SC, LU	TR or IN	Bulk shipping	RO	D
Cho and Perakis (1996)	P, B, CI, LU	LI	General shipping	MIP	D
Jin and Kite-Powell (2000)	B, SC	TR, LI or IN	General shipping	OC	–
Meng and Wang (2011)	P, CI, CO, SE, LU	LI	Container shipping	DP	D
Nicholson and Pullen (1971)	CI, SE	TR, LI or IN	General shipping	DP	–
Wijsmuller and Beumee (1979)	B, CI, SE, LU	LI	General shipping	LP	–
Xie et al. (2000)	P, LU	TR, LI or IN	General shipping	DP	D

the fleet size and mix on a tactical level only. Table 2 summarizes the ones belonging to the former category.

The first contribution to this category of problems was given by Everett et al. (1972), who presented the problem of composing the US merchant marine fleet. A linear programming (LP) problem was used to obtain the best ship designs and sizes for a fleet of tanker and bulkers supposed to carry 15% of the US foreign trade (e.g. grain, oil, coal and ores). The problem was later slightly modified and re-proposed by Bradley et al. (1977). During the same period, Murotsu and Taguchi (1975) combined dynamic programming (DP) and non-linear programming to determine the optimal fleet size and mix for a fleet of crude oil carriers. They assumed only one origin port and one destination port. The characteristics of the ships (size and speed) were also variables of the problem and hence determined by their procedure instead of being fixed in advance.

Later, a number of authors proposed heuristic procedures for the solution of the problems they illustrated. Larson (1988) studied the problem of transporting sludge from waste water treatment plants in the city of New York to an offshore dumping site. Among other strategic issues, the number and the size of the vessels to employ were to be determined. Pesenti (1995) proposed a heuristic procedure to support purchasing and use of container ships. The algorithm consists of exchange of information and decisions between all planning levels. Strategic decisions are passed to the tactical level to be tested and receive a feedback. Then the strategic

level can amend its solution or make it final. Sigurd et al. (2005) studied the problem of establishing a new liner shipping system for container transportation from Norway to Central Europe. They evaluated the possibility of building up to 15 different ships to ensure the desired service speed and frequency. The problem was formulated as a set partitioning problem and solved by means of a heuristic branch-and-price algorithm. Zeng and Yang (2007) described the problem of deciding both fleet size and mix and ship schedules for a Chinese coal shipping system. Coal was transported by rail to three outbound ports and then shipped to demand ports. A tabu search heuristic was used to solve the original integer programming (IP) formulation of the problem. Crary et al. (2002) aimed to determine the US destroyers fleet size and mix in sight of a potential conflict on the Korean peninsula. Each mission had a random weight dependent on the actual war dynamics. They solved multiple times a mixed integer programming (MIP) problem under different realization of the weights of the missions in order to estimate the war winning probability and to find the fleet performing the best.

Two papers describe the use of simulation to evaluate alternative fleets. The first one, proposed by Darzentas and Spyrou (1996), treats the development of a ferry system in the Aegean Sea. They simulated the ferry traffic under different settings of the system (e.g. port layouts and ferry routes), including different ferry fleet configurations. Key indicators were for example the passenger's delay and the number of ships in queue at ports. A similar

approach was used by Fagerholt et al. (2010). A set of different strategic decisions were given as input to a simulation tool. The tool first draws realizations for a set of elements not under the company's control (e.g. cargo quantities and time windows) and then solves short term ship routing and scheduling problems during the simulation. The set of strategic decisions may include fleet size and mix decisions, although the case study presented did not.

In the short-term MFSMP a fleet of ships needs to be arranged to fulfill a short-term task. Typically, ships are chosen from an available pool (e.g. a pool of ships shared between more transportation tasks) or chartered in for the length of the transportation task. From this it is understood that these are not strategic problems. Table 3 summarizes the papers on the short-term MFSMP.

To the best of our knowledge, the earliest paper in this category belongs to Schwartz (1968). A bargeline company is to move given cargoes from origins to destinations in its district at given times. The problem consists of determining the number of barges and towboats of each size to move the cargoes. Barges and boats can be chosen from an available pool. The problem is modeled by means of an IP problem but it could not be solved with the algorithms available at that time. In the paper proposed by Mehrez et al. (1995), bulk products are to be shipped to transshipment ports along the Atlantic Coast and the decision maker is to decide the number and the type of ships to charter in each time period to perform the shipments. They developed a heuristic algorithm to solve the IP model proposed for the problem. Meng and Wang (2010) presented a short-term MFSMP for a container liner shipping company that consists of deciding which of the available ships to use and their deployment as well as the number of charters (in or out). They proposed a chance constrained problem to tackle demand uncertainty. Lately, Meng et al. (2012) extended this problem including transshipment. They proposed a two-stage stochastic IP model as well as a solution algorithm.

Fagerholt and Lindstad (2000) studied the supply service for an offshore installation with the scope of finding which vessels to operate (charter) on weekly schedules. They first generated a number of feasible ship routes and then, by solving an IP problem, chose the schedule and the corresponding vessels to use. Later, Halvorsen-Weare et al. (2012) studied a similar problem including more realistic elements such as spreads of departures and maximum and minimum duration of voyages. Their optimization model was then used by Halvorsen-Weare and Fagerholt (2011) in combination with simulation to ensure more robust routes and fleet solutions with respect to the weather. A liner shipping service in the Pacific Sea was considered by Lane et al. (1987). A MIP problem selects among a set of candidate ship schedules found by a forward-looking heuristic. Ship schedules are associated with ships, therefore the fleet size and mix emerges from the selection of schedules. A similar approach was used by Fagerholt (1999). The context was that of the development of a new liner shipping system from the Norwegian Sea to ports in Europe and US. The focus was on finding the optimal fleet of feeder vessels to hire for fixed periods in order to move cargoes from Norway to a central depot. Complete ship routes are generated and then a set partitioning problem selects the most convenient routes and the associated vessels.

3.2. Maritime fleet size problems

The MFSP is a special case of MFSMP, in which there is only one ship type. Hence, the problem is to decide the number of ships required to perform some transportation task. Table 4 summarizes the research papers related to the MFSP.

The first contribution on MFSPs dates back to 1954 (Dantzig and Fulkerson, 1954). The authors applied an LP model to determine the minimum number of tankers in order to guarantee a shipping

service with a fixed schedule. Later, Bellmore et al. (1968) modified this problem by including a utility for each delivery and assessing the possibility to cancel some deliveries by giving up the corresponding utility. An LP model was used in that paper as well.

Bendall and Stent (2001) presented the problem of finding the number of identical container ships to be assigned to a high speed service in a major hub and spoke system based in Singapore. The MIP model also provided optimal deployment for the ships. The number of ferries was to be chosen in the framework of ferry network design studied by Lai and Lo (2004). The objective was that of finding an optimal ferry network configuration taking into account set up and operating costs as well as passengers' performance measures (e.g. waiting time). They used a heuristic algorithm to solve the MIP formulation for the problem. Sambracos et al. (2003) investigated the introduction of small containers in a coastal shipping system in the Aegean Sea. They proposed a MIP model to determine the number of similar container vessels to use while ship operations were modeled as a vehicle routing problem.

Two studies dealt with barge-tugs systems. Jaikumar and Solomon (1987) studied the minimization of the number of tugs to acquire in order to move a given number of barges between ports in a river system. By exploiting the particular structure of the problem they proposed a polynomial exact algorithm to solve the IP formulation of the problem. On commission of the New York City department of sanitation Larson et al. (1991) developed a simulation tool for supporting the task of determining the number of barges to transport refuse to an offshore landfill. The performance of different fleet sizes was tested for example in terms of tonnage of delayed refuse and time spent at the dumping site. This work was later extended by Richetta and Larson (1997) to adapt the simulation tool to the changes occurred in the marine transport system due to new environmental regulations and technologies.

Some other studies described the use of simulation. Shyshou et al. (2010) dealt with supply vessels for anchor handling operations. Such vessels are hired either on long-term charters or on spot charters which are usually more expensive. The objective was that of finding the cost optimal fleet of vessels on long-term charter. Different fleet sizes were evaluated in a discrete event simulation framework. A fleet of liquid natural gas vessels was to be assembled in the papers of Koenigsberg and Lam (1976) and Koenigsberg and Meyers (1980). The former considered gas transportation between one loading and one or two discharge ports while the latter applied the model to the transportation of gas between Alaska and Los Angeles. In both papers different fleet sizes were evaluated in a simulation tool and the underlying operations were described by means of queueing models.

3.3. Maritime fleet renewal problems

In this section we report on the available papers on the MFRP. The papers reviewed are summarized in Table 5.

Xie et al. (2000) studied the development of the fleet over time in order to meet a given demand pattern. In every year ships can be added to the initial fleet or laid up. The fleet development was modeled by means of DP while the underlying fleet deployment was found by solving LP problems. A similar structure was employed by Meng and Wang (2011) who developed a multi-period fleet planning model for liner shipping companies. Also in this case DP is applied to model the fleet evolution but here the fleet deployment is found by solving a MIP problem for each time period. In the DP model each status represents one of a number of scenarios proposed by experts, where a scenario consists of the number of ships owned, bought, chartered in or chartered out in the corresponding period of time.

Some papers aimed at finding the best replacement schedule for the available ships while also allowing the fleet size and mix to

vary between periods. Wijsmuller and Beumee (1979) presented an LP model for a ship investment and replacement problem. The fleet size could vary within an upper and a lower bound and also the fleet mix could be adjusted while finding the replacement schedule. Jin and Kite-Powell (2000) aimed at determining optimal strategies for using, building and scrapping ships in order to maximize the profit. The number of vessels was allowed to vary over time and ships could be bought or scrapped not only for replacement purposes. They employed optimal control theory for its formulation and solution. Nicholson and Pullen (1971) studied the problem of scaling down a fleet of ships due to major technology changes. In this problem the size of the fleet is given (i.e. the number of ships to use in each period) but the mix of owned and chartered ships must be determined. Charters are adopted in case of premature sale of available ships. A heuristic procedure schedules the sales of the ships while the optimal number of charters is found by means of DP.

Cho and Perakis (1996) studied the problem of expanding a current liner shipping fleet by evaluating a set of ships to include (build, purchase or charter). They did not run experiments but suggested a Lagrangian relaxation for solving the MIP formulation proposed for the problem. At last, Alvarez et al. (2011) explicitly considered uncertainty while studying the fleet evolution. A robust optimization model was proposed to find solutions which are feasible against random variations in the selling and purchasing prices of ships. Their method is suitable for companies with varying degrees of risk tolerance in managing and modifying the fleet. In order to solve the model, it was transformed into a MIP problem.

3.4. Fleet adjustment in tactical problems

At a tactical level of planning a fleet of ships is given and the focus is on the best way of using the available ships in order to meet the transportation demand. Typical problems faced at this level are fleet deployment and ship routing and scheduling problems. The former consists of finding the minimum cost allocation of the available ships to services (e.g. trades and routes). Generally, the decision is about the number of times each ship (or type of ships) will operate a given service. The latter is usually faced in tramp and industrial shipping operations and consists of sequencing the ports to call (ship routing) and sometimes of assigning a time to each port call (ship routing and scheduling).

In most cases, when the fleet size and mix is planned (that is an MFSMP is solved), the actual realization of the demand is not known for sure. Therefore, in a tactical planning context, given the actual demand, decision makers evaluate the need to charter in additional ships. Similarly, if the transportation demand can be met by a subset of the available ships, decision makers can decide to charter out ships to other companies or just lay them up at port.

Numerous studies include the possibilities to charter in or out or lay up ships when dealing with tactical planning. We limit our-

selves to report a few examples on how the fleet is adjusted within the frame of tactical problems. A more thorough discussion of the literature on fleet deployment and ship routing and scheduling problems is provided by Christiansen et al. (2004) and Christiansen et al. (2007).

Table 6 summarizes the papers we include here. We report the main tactical problem studied in the paper, the type of action that modifies the fleet size and mix (e.g. charter in or charter out), the operations mode and the shipping industry dealt with.

The most common alternative to adjust the fleet is the lay up. It consists of excluding ships from operations for a period of time by stopping them at port with reduced crew and engines kept at least (safety) regimes. The operating costs of the ships laid up are therefore reduced during this period. Perakis and Jaramillo (1991) and Jaramillo and Perakis (1991) introduced lay up possibilities in a fleet deployment problem. The former presented an LP formulation for the problem while the latter presented its implementation and results. Later, this work was modified by Powell and Perakis (1997), who introduced an IP model to eliminate rounding errors likely to occur when using the LP formulation of the former authors.

Charters (in or out) and lay up periods are included in the fleet deployment problem studied by Gelareh and Meng (2010). Their MIP model was later reformulated by Wang et al. (2011) in order to eliminate its combinatorial behavior, while removing lay up options. Finally, Wang and Meng (2012) also considered only charter in and charter out possibilities to correct the fleet structure. The problem they proposed is a fleet deployment problem for a liner shipping company and includes handling of transshipment operations.

Fleet modifications have also been treated in ship routing and scheduling problems. Rana and Vickson (1988) provided an IP model to help liner shipping companies evaluating whether individual container ships should be chartered in or not within the framework of a ship routing problem. An exact specialized algorithm was proposed to solve the problem. Sherali et al. (1999) included time charter options in a ship routing and scheduling problem where tankers are transporting oil from Kuwait to Asia, America, and Europe. The MIP model presented is initially reformulated in a more tractable model and then solved by means of a specialized rolling horizon heuristic algorithm. Hwang et al. (2008) included the possibility to charter out some of the available ships when capacity surplus occurs. They presented a quadratic model to limit the variance of shipping profit caused by volatility in the spot market. The model is solved by means of a Branch-and-Price-and-Cut algorithm.

4. Research contributions and future perspectives

The first element we point out is the general scarcity of papers dealing with the maritime fleet size and mix problem (MFSMP) and its variants presented in Sections 3.1, 3.2, 3.3. We count 37 papers

Table 6
Papers including tactical corrections to the fleet size and mix.

Paper	Operating decisions	Acquisition/disposal	Mode	Industry
Gelareh and Meng (2010)	D	CI, CO, LU	LI	General shipping
Hwang et al. (2008)	S	CO	TR or LI	General shipping
Jaramillo and Perakis (1991)	D	LU	LI	General shipping
Perakis and Jaramillo (1991)	D	LU	LI	General shipping
Powell and Perakis (1997)	D	LU	LI	General shipping
Rana and Vickson (1988)	R	CI	LI	Container shipping
Sherali et al. (1999)	R + S	CI	IN	Oil shipping
Wang et al. (2011)	D	CI, CO	LI	General shipping
Wang and Meng (2012)	D	CI, CO	LI	Container shipping

produced from 1954 to 2012. This number is small if compared to the 60 papers mostly on ship routing and scheduling published only in the decade before 2004 and reviewed by Christiansen et al. (2004). Possible motivations to this shortage of papers might be: (1) the difficulty to plan for long periods of time and thus to have access to market information which in most cases is imprecise or even not available, (2) the ocean shipping industry, due to its long tradition, is conservative, especially with regard to investment decisions which are typically made based on past experience, entrepreneurial spirit, and taking advantage of different national laws (Ronen, 1983). However, the frequent tonnage imbalances discussed in Section 1 and the scarcity of papers suggest more effort to improve decision support systems and make the MFSMP an open field for new research initiatives.

Most of the papers reviewed studied single-period problems (see Section 3.1.). The methods proposed fit shipping contexts which are meant to be stable or short-term problems where changes and uncertainties are cut off by the limited planning horizon. The research literature covers this kind of problems with a number of different analytic methodologies and simulation models. Furthermore, the available papers address or can fit the start-up of completely new shipping systems. They in fact implicitly do not assume an existing fleet.

However, given the dynamic nature of the shipping market and the long lifetime of ships, most real life problems need to take into account the evolution of the market status rather than a single, unchangeable, period. Furthermore, only in rare occasions a completely new fleet needs to be determined. More often shipping companies need to adjust the current fleet according to the market evolution and their beliefs about the future. Only in seven of the 37 reviewed papers, the authors assume that there is an initial fleet that should be adjusted, over time, by including or excluding ships (see Section 3.3).

Moreover, 10 of the 37 papers address the dimensioning of a homogeneous fleet (see Section 3.2.). As far as ocean transportation is concerned more often than not fleets are heterogeneous, varying at least in size, technology, cost function and operating restrictions.

Additional research should be encouraged to capture an increasing number of realistic elements of the problem. This could be done by better addressing the process of having a step-by-step evolution of the fleet in response to the market dynamics rather than a static configuration of the fleet. The emphasis should be on the best modification of the current fleet, taking into account future market challenges. This effort needs to consider the heterogeneous characteristics of ocean going fleets and a possible challenge could be the definition of ship types and the evaluation of the effects of different clustering strategies (i.e. how ships are grouped by similar characteristics).

Most reviewed papers propose methods for planning in a deterministic context. Although some papers include stochastic elements – see Table 7 – in most cases they still propose methods that are fitted for deterministic problems. Uncertain data is typically replaced by average or extreme values. As an example, in Crary et al. (2002) the importance (weight) of each mission of a potential war is stochastic. They test different fleets with different realizations of the random weights of the missions. Each test consists of solving a deterministic MIP to estimate the war winning probability. After a number of trials the best performing fleet is chosen. Another example is given by Jaikumar and Solomon (1987). When they incorporate stochastic demand they solve an IP problem planning for the highest value of demand.

Decision problems which can be reasonably considered deterministic are rather scarce in maritime transportation. This is especially true for strategic planning problems with a long planning horizon such as the strategic MFSMP. Only few papers explicitly treat uncertainty. If only strategic problems are considered the only

Table 7

Papers including uncertain elements in the problem.

Paper	Problem	Planning against uncertainty	Simulation
Alvarez et al. (2011)	MFRP	x	–
Crary et al. (2002)	MFSMP	–	–
Fagerholt et al. (2010)	MFSMP	–	x
Halvorsen-Weare and Fagerholt (2011)	MFSMP	–	x
Jaikumar and Solomon (1987)	MFSMP	–	–
Larson et al. (1991)	MFSMP	–	–
Meng et al. (2012)	MFSMP	x	–
Meng and Wang (2010)	MFSMP	x	–
Pesenti (1995)	MFSMP	–	–
Shyshou et al. (2010)	MFSMP	–	x
# of papers	10	3	3
% of the 37 papers reviewed	27	8.1	8.1

exception is the robust optimization model presented by Alvarez et al. (2011) (see Section 3.3). Fagerholt et al. (2010), and Shyshou et al. (2010) faced uncertainty by means of simulation tools. The former included uncertainty in the quantity and timing of the cargoes while the latter considered uncertain weather conditions in offshore operations. Finally, Halvorsen-Weare and Fagerholt (2011), Meng and Wang (2010) and Meng et al. (2012) included and tackled uncertainty in short-term MFSMPs (see Section 3.1).

Future research needs to increase the attention towards more sophisticated methods to handle uncertainty or at least better investigate in which cases deterministic programming approaches may perform well. A possible challenge can be the modeling of the complex shipping market behavior.

Although in this paper we exclude a thorough examination of papers modeling ships as marketable assets, studying the MFSMP under the perspective of asset play remains a useful avenue for future research. The MFSMP can be seen within the more general frame of the shipper's capital and investment decisions. The combination of operations research techniques and real option analysis might represent a valuable tool for this purpose.

As far as the cost function is concerned the most common modeling approach is that of incurring a fixed cost when adding a ship to the fleet and a variable cost associated with the sailing of the ships. The fixed cost is described in more or less detail from one study to another, and a number of them do not specify its meaning. However, in almost all cases the cost is univocal and no comparison between different alternatives is given. Real life problems typically offer a wealth of options when it comes to include or exclude ships from the fleet. Ships can be built, bought in the second-hand market, chartered in or taken as bare boat charter (i.e. the ship is chartered for a given period at an agreed rate, the owning company pays the capital cost and the charterer all other costs). Each of these options is associated with different needs and outcomes. As an example, second-hand ships are available in short time, but their characteristics must be chosen among what is available in the market. On the opposite, newbuildings can be designed according to the company's specifications, but are available after several months to years. Similarly, when ships are to be disposed of a shipping company can charter them out, lay them up, scrap or sell them. Therefore, a potential venue for research is to better address the type of fleet modification by comparing different acquisition or disposal options.

Almost all the reviewed papers use the underlying ship operation to estimate the total capacity needed to meet the demand. However, different levels of detail in the routing aspects can be found in the literature. Tables 2–5 report the operating decisions on which the tonnage estimation of the reported papers is based. Operating decisions vary from ship scheduling to ship deployment.

The former alternative implies much more information (e.g. the exact sequencing of port calls and their timing) while the latter is at a higher level of abstraction. Open questions may be how different levels of detail in modeling the operations of the ships influence the resulting fleet size and mix decisions, and what is the least amount of information needed to make sound strategic decisions. The answer to these questions will surely vary between different shipping segments and operation modes. Future research may set out to explore and better describe these aspects.

Finally, with regard to computation issues, difficulties arise in making a fair comparison between the methodologies proposed in the literature. The first obstacle is the wealth of different shipping networks and fleet operations described which makes it hard to evaluate and compare the complexity of the different problems. However, a significant number of papers did not apply heuristic procedures (see Tables 2–5), solving the problems to optimality. A possible interpretation of this is that the field is ready to add extra complexity by including more real life elements. On the other hand, standardized benchmark problems are needed in order to allow a more thorough evaluation of the computational efficiency. This can in turn trigger new research initiatives on better solution methods and on variants of the problem obtained by including additional problem features.

5. Conclusions

Due to the changeable and stochastic behavior of the seaborne economy, decisions on the composition of the fleet of ships are crucial in maritime transportation. Furthermore, frequent tonnage imbalances call for better decision support systems. We surveyed the available literature on the maritime fleet size and mix problem and discussed the state of the art on the decision support for such problem. We pointed out the available methodologies and the characteristics of the problems they are devised for. Furthermore, we highlighted points of weakness, where the available methodologies fail to capture important real life aspects of the problem and suggested possible future research directions to enhance the state of the art.

The available research papers have been focusing on shipping systems which can be considered as static and not characterized by volatility. They have also well addressed the start-up of new shipping systems. However, important aspects of real life problems deserve further attention. Future research should especially focus more on the renewal of the fleet where the continual adaption of the fleet to changes in the market situation is sought. Important characteristics of the problem to better take into account are the uncertain market behavior and the large number of alternative ways for the shipping companies to renew their fleet. Furthermore, open research questions can be found, such as the appropriate description of the operations of the ships and opportune methodologies to include and tackle uncertainty.

Acknowledgments

We want to thank Professor Christian Bierwirth for the precious suggestions about an earlier version of the paper. Thanks also to the two anonymous referees for the insightful comments. This work was supported by the MARFLIX and DOMinant II projects, partly funded by the Research Council of Norway

References

Adland, A.O.S., Koekebakker, S., 2004. Market efficiency in the second-hand market for bulk ships. *Maritime Economics & Logistics* 6, 1–15.
Adland, R., Koekebakker, S., 2007. Ship valuation using cross-sectional sales data: a multivariate non-parametric approach. *Maritime Economics & Logistics* 9, 105–118.

Alizadeh, A.H., Nomikos, N.K., 2007. Investment timing and trading strategies in the sale and purchase market for ships. *Transportation Research Part B: Methodological* 41 (1), 126–143.
Alvarez, J.F., Tsilingiris, P., Engbrethsen, E.S., Kakalis, N.M.P., 2011. Robust fleet sizing and deployment for industrial and independent bulk ocean shipping companies. *INFOR: Information Systems and Operational Research* 49, 93–107.
Bellmore, M., Bennington, G., Lubore, S., 1968. A maximum utility solution to a vehicle constrained tanker scheduling problem. *Naval Research Logistics Quarterly* 15, 403–412.
Bendall, H.B., Stent, A.F., 2001. A scheduling model for a high speed containership service: a hub and spoke short-sea application. *International Journal of Maritime Economics* 3, 262–277.
Bendall, H.B., Stent, A.F., 2005. Ship investment under uncertainty: valuing a real option on the maximum of several strategies. *Maritime Economics & Logistics* 7, 19–35.
Bradley, S.P., Hax, A.C., Magnanti, T.L., 1977. Planning the mission and composition of the US merchant marine fleet. In: *Applied Mathematical Programming*. Addison-Wesley (Chapter 7).
Cho, S.C., Perakis, A.N., 1996. Optimal liner fleet routeing strategies. *Maritime Policy & Management* 23, 249–259.
Christiansen, M., Fagerholt, K., Ronen, D., 2004. Ship routing and scheduling: status and perspectives. *Transportation Science* 38, 1–18.
Christiansen, M., Fagerholt, K., Ronen, D., Nygreen, B., 2007. Maritime transportation. In: Barnhart, C., Laporte, G. (Eds.), *Handbook in Operations Research and Management Science*. Elsevier, pp. 189–284.
Couillard, J., Martel, A., 1990. Vehicle fleet planning the road transportation industry. *IEEE Transactions on Engineering Management* 37, 31–36.
Crary, M., Nozick, L.K., Whitaker, L.R., 2002. Sizing the US destroyer fleet. *European Journal of Operational Research* 136, 680–695.
Dantzig, G., Fulkerson, D.R., 1954. Minimizing the number of tankers to meet a fixed schedule. *Naval Research Logistics Quarterly* 1, 217–222.
Darzentas, J., Spyrou, T., 1996. Ferry traffic in the Aegean islands: a simulation study. *The Journal of the Operational Research Society* 47, 203–216.
Everett, J.L., Hax, A.C., Lewinson, V.A., Nudds, D., 1972. Optimization of a fleet of large tankers and bulkers: a linear programming approach. *Marine Technology* (October), 430–438.
Fagerholt, K., 1999. Optimal fleet design in a ship routing problem. *International Transaction in Operational Research* 6, 453–464.
Fagerholt, K., Christiansen, M., Hvattum, L.M., Johnsen, T.A.V., Vabø, T.J., 2010. A decision support methodology for strategic planning in maritime transportation. *Omega* 38, 465–474.
Fagerholt, K., Lindstad, H., 2000. Optimal policies for maintaining a supply service in the Norwegian sea. *Omega* 28, 269–275.
Gelareh, S., Meng, Q., 2010. A novel modeling approach for the fleet deployment problem within a short-term planning horizon. *Transportation Research Part E: Logistics and Transportation Review* 46, 76–89.
Halvorsen-Weare, E.E., Fagerholt, K., 2011. Robust supply vessel planning. In: *Network Optimization*. In: Pahl, J., Reiners, T., Voß, S. (Eds.), *Lecture Notes in Computer Science*, vol. 6701. Springer, Berlin/Heidelberg, pp. 559–573.
Halvorsen-Weare, E.E., Fagerholt, K., Nonås, L.M., Asbjørnslett, B.E., 2012. Optimal fleet composition and periodic routing of offshore supply vessels. *European Journal of Operational Research* 223, 508–517.
Hoff, A., Andersson, H., Christiansen, M., Hasle, G., Løkketangen, A., 2010. Industrial aspects and literature survey: fleet composition and routing. *Computers & Operations Research* 37, 2041–2061.
Hwang, H.S., Visdildilokpun, S., Rosenberg, J.M., 2008. A branch-and-price-and-cut method for ship scheduling with limited risk. *Transportation Science* 42, 336–351.
Jaikumar, R., Solomon, M.M., 1987. The tug fleet size problem for barge line operations: a polynomial algorithm. *Transportation Science* 21, 264–272.
Jaramillo, D., Perakis, A., 1991. Fleet deployment optimization for linear shipping part 2. Implementation and results. *Maritime Policy & Management* 18, 235–262.
Jin, D., Kite-Powell, H.L., 2000. Optimal fleet utilization and replacement. *Transportation Research Part E* 36, 3–20.
Koenigsberg, E., Lam, R.C., 1976. Cycling queue models for fleet operations. *Operations Research* 24, 516–529.
Koenigsberg, E., Meyers, D.A., 1980. An interacting cyclic queue model of fleet operations. *The Logistics and Transportation Review* 16, 59–71.
Lai, M.F., Lo, H.K., 2004. Ferry service network design: optimal fleet size, routing, and scheduling. *Transportation Research Part A* 38, 305–328.
Lane, D.E., Heaver, T.D., Uyeno, D., 1987. Planning and scheduling for efficiency in liner shipping. *Maritime Policy & Management* 14, 109–125.
Larson, R.C., 1988. Transporting sludge to the 106-mile site: an inventory/routing model for fleet sizing and logistics system design. *Transportation Science* 22, 186–198.
Larson, R.C., Minkoff, A., Gregory, P., 1991. A computer simulation model for fleet sizing for the marine division of the New York City department of sanitation. *Waste Management & Research* 9, 267–276.
Lawrence, S.A., 1972. *International Sea Transport: The Years Ahead*. Lexington Books, Lexington.
Marcus, H.S., Glucksman, M.A., Ziogas, B.O., Meyer, K.L., 1991. A buy-low, sell-high investment methodology: the case of bulk shipping. *Interfaces* 21, 8–21.
Mehrez, A., Hung, M.S., Ahn, B.H., 1995. An industrial ocean-cargo shipping problem. *Decision Sciences* 26, 395–423.
Meng, Q., Wang, T., 2010. A chance constrained programming model for short-term liner ship fleet planning problems. *Maritime Policy & Management* 37, 329–346.

- Meng, Q., Wang, T., 2011. A scenario-based dynamic programming model for multi-period liner ship fleet planning. *Transportation Research Part E* 47, 401–413.
- Meng, Q., Wang, T., Wang, S., 2012. Short-term liner ship fleet planning with container transshipment and uncertain container shipment demand. *European Journal of Operational Research* 223, 96–105.
- Murotsu, Y., Taguchi, K., 1975. Optimization of ship fleet-size. *Bulletin of University of Osaka Prefecture, Series A: Engineering and Natural Sciences* 23, 171–192.
- Nicholson, T.A.J., Pullen, R.D., 1971. Dynamic programming applied to ship fleet management. *Operational Research Quarterly* 22 (1970–1977), 211–220.
- Perakis, A., Jaramillo, D., 1991. Fleet deployment optimization for liner shipping part 1. Background, problem formulations and solution approaches. *Maritime Policy & Management* 18, 183–200.
- Pesenti, R., 1995. Hierarchical resource planning for shipping companies. *European Journal of Operational Research* 86, 91–102.
- Powell, B., Perakis, A., 1997. Fleet deployment optimization for liner shipping: an integer programming model. *Maritime Policy & Management* 24, 183–192.
- Rana, K., Vickson, R.G., 1988. A model and solution algorithm for optimal routing of a time-chartered containership. *Transportation Science* 22, 83–85.
- Richetta, O., Larson, R.C., 1997. Modeling the increase complexity of New York City's refuse marine transport system. *Transportation Science* 31, 272–293.
- Ronen, D., 1983. Cargo ships routing and scheduling: survey of models and problems. *European Journal of Operational Research* 12, 119–126.
- Sambracos, E., Paravantis, J.A., Tarantilis, C.D., Kiranoudis, C.T., 2003. Dispatching of small containers via coastal freight liners: the case of the Aegean Sea. *European Journal of Operational Research* 152, 365–381.
- Schwartz, N.L., 1968. Discrete programs for moving known cargos from origins to destinations on time at minimum barge fleet cost. *Transportation Science* 2, 134–145.
- Sherali, H.D., Al-Yakoob, S.M., Hassan, M.M., 1999. Fleet management models and algorithms for an oil-tanker routing and scheduling problem. *IIIE Transactions* 31, 395–406.
- Shyshou, A., Gribkovskaia, I., Barceln++, J., 2010. A simulation study of the fleet sizing problem arising in offshore anchor handling operations. *European Journal of Operational Research* 203, 230–240.
- Sigurd, M.M., Ulstein, N.L., Nygreen, B., Ryan, D.M., 2005. Ship scheduling with recurring visits and visit separation requirements. In: *Column Generation*. Springer, pp. 225–245.
- Sødal, S., Koekebakker, S., Adland, R., 2009. Value based trading of real assets in shipping under stochastic freight rates. *Applied Economics* 41 (22), 2793–2807.
- Stopford, M., 2009. *Maritime Economics*. Routledge, New York.
- Tsolakis, S.D., Cridland, C., Haralambides, H.E., 2003. Econometric modelling of second-hand ship prices. *Maritime Economics & Logistics* 5, 347–377.
- UNCTAD, 2010. *Review of maritime transport, 2010*. United Nations, New York and Geneva.
- UNCTAD, 2011. *Review of maritime transport, 2011*. United Nations, New York and Geneva.
- Wang, S., Meng, Q., 2012. Liner ship fleet deployment with container transshipment operations. *Transportation Research Part E: Logistics and Transportation Review* 48, 470–484.
- Wang, S., Wang, T., Meng, Q., 2011. A note on liner ship fleet deployment. *Flexible Services and Manufacturing Journal* 23, 422–430.
- Wijsmuller, M.A., Beumee, J.G.B., 1979. Investment and replacement analysis in shipping. *International Shipbuilding Progress* 26, 322–343.
- Xie, X., Wang, T., Chen, D., 2000. A dynamic model and algorithm for fleet planning. *Maritime Policy & Management* 27, 53–63.
- Zeng, Q., Yang, Z., 2007. Model integrating fleet design and ship routing problems for coal shipping. *Lecture Notes in Computer Science* 4489, 1000–1003.